Quiz#6

Section: _____ **Group:** _____

Name(print): _____ **Student ID:** _____

1. If the magnitudes of both charges are tripled and their separation is also tripled, what happens to the magnitude of the electrostatic force that each charge exerts on the other?

   A. It stays the same.
   B. It decreases by a factor of $3^{1/2}$.
   C. It increases by a factor of 3.
   D. It increases by a factor of 9.
   E. It increases by a factor of $3^{1/2}$.

2. A first charge, $q_1 = -12\mu C$, is located at $\vec{r}_1 = (-4.5\text{m})\hat{\jmath}$. A second charge, $q_2 = -37\mu C$, is located at $\vec{r}_2 = (+8.3\text{m})\hat{\jmath}$.
   a. Draw a diagram that includes the two charges, q_1 and q_2, within a 2D coordinate system showing their positions. Additionally, represent the electric force exerted by q_1 on q_2 as a vector in the diagram.
   b. Determine the x- and y-component of the electric force on q_2.